

RAGURAM KASTHURISAMY

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EDUCATION

Northeastern University <i>Master of Science in Electrical and Computer Engineering</i> Coursework: Computer Architecture, Computer Systems	<i>Boston, MA</i> Sep 2026 - Dec 2028 (Expected)
Kumaraguru College of Technology <i>Bachelor of Engineering in Mechatronics Engineering</i> Coursework: Introduction to Internet of Things (IoT), Advanced Internet of Things (IoT), Digital Electronics & Microprocessors, Embedded Systems, Industrial Electronics & Drives	<i>Coimbatore, India</i> 2020 - 2024 GPA: 9.14/10

EXPERIENCE

National Instruments (Contract) <i>Project Engineer</i> - Built software to monitor, analyze, and manage distributed measurement systems, delivering real-time telemetry and system monitoring for production users - Developed automated regression test procedures and tools achieving 100% test code coverage, validating performance and ensuring reliable, defect-free operation - Collaborated across cross-functional engineering teams to deliver customer-critical workflows under tight deadlines	<i>Coimbatore, India</i> Jan 2026 - Aug 2026
KLA India (Contract) <i>Project Engineer</i> - Developed a C# .NET application to measure and characterize overlay in printed wafers for semiconductor fabrication metrology - Drove root-cause resolution of release-blocking bugs each cycle and integrated static code analysis into the Jenkins CI/CD pipeline, reducing memory usage by 50%	<i>Chennai, India</i> Sep 2024 - Dec 2025
Soliton Technologies Pvt. Ltd. <i>Intern</i> - Contributed to the Protocol Validation Suite (PVS) for I3C and SPI, developing and running tests to validate communication-protocol implementations against specification - Built automation and tooling to support protocol conformance testing, debugging, and result reporting	<i>Coimbatore, India</i> Jan 2024 - Aug 2024
iQube - Innovation Forum, Kumaraguru College of Technology <i>Embedded and IoT Intern</i> - Developed low-level embedded C/C++ firmware for AVR, STM32 (ARM Cortex-M), and PIC32 — device drivers, hardware interfaces, interrupts, state machines, and fault handling - Debugged issues at the firmware/hardware boundary across digital and analog hardware using oscilloscopes and logic analyzers to root-cause FW/HW faults - Performed hardware bring-up and built communication over SPI, I2C, UART, and CAN; read schematics and datasheets to resolve integration issues - Built home-automation and industrial IoT solutions using Arduino (Uno/Nano), ESP32, ESP32-CAM, ESP8266, and Raspberry Pi, integrating sensors and actuators for real-world monitoring and control - Established multi-device communication between ESP32, Raspberry Pi, and Arduino nodes in master-slave configurations, enabling coordinated data collection and control across networked devices	<i>Coimbatore, India</i> Mar 2021 - Dec 2023

TECHNICAL SKILLS

Firmware & Low-Level	Embedded C/C++, ARM Cortex-M (STM32), AVR, PIC32, bare-metal & RTOS-based systems, bootloaders & boot/system init, low-level device drivers, embedded controller structure
Hardware & Debug	Digital & analog hardware, PCB design (KiCad, Altium), hardware bring-up, schematics & datasheets; oscilloscopes, logic analyzers, JTAG / GDB debuggers, root-cause analysis
IoT & Platforms	Arduino (Uno, Nano), ESP32, ESP32-CAM, ESP8266, Raspberry Pi; home automation, industrial IoT solutions, sensor & actuator integration, master-slave device communication, Wi-Fi connectivity, edge prototyping
Architecture & Buses	Firmware & hardware design principles, computer architecture; SPI, I2C, UART, CAN, ADC/DAC; interrupts & timing
Languages & Tools	C, C++, Python, C#/.NET, shell scripting; Git / GitHub, Jira, compilers, IDEs, Windows & Linux
Test & Process	Test plans & protocols, unit / integration testing, 100% code coverage, Jenkins CI, static analysis, code reviews, Agile, documentation
AI & Extras	Prompt engineering, LLM-assisted debugging, RAG systems

PROJECTS

Custom STM32 Bootloader - Developed a custom STM32 bootloader in C/C++ enabling in-field firmware updates over UART and SPI, with flash memory partitioning and safe jump-to-application logic - Implemented CRC/checksum validation and a fault-tolerant recovery state machine to prevent device bricking on failed or corrupted updates - Designed a packet-based transfer protocol with error handling and retransmission for reliable firmware delivery
Real-Time Task Scheduler Built a cooperative real-time scheduler for PIC32 in C, implementing deterministic, timing-sensitive task scheduling on bare metal without RTOS overhead

PUBLICATIONS & ACHIEVEMENTS

<i>An Automated Mechanized Robot for Handling Rice Bags</i> - Intl. Journal for Multidisciplinary Research (2024) - Best Outgoing Student, Mechatronics Dept. (2024) Winner, Smart India Hackathon (2022) CTO, iQube Forum, KCT
